

Wind Physics Measurement Project

Lidar module

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SPECTRA CLEANING

- Each spectrum sits on a **noise floor $\approx 5,000$ a.u.**
- Doppler peak at bin $\approx 65,000$ a.u.
- Applied **Simple Median Subtraction** per spectrum.

Median ignores the peak.

[Why does Median ignore the peak?]

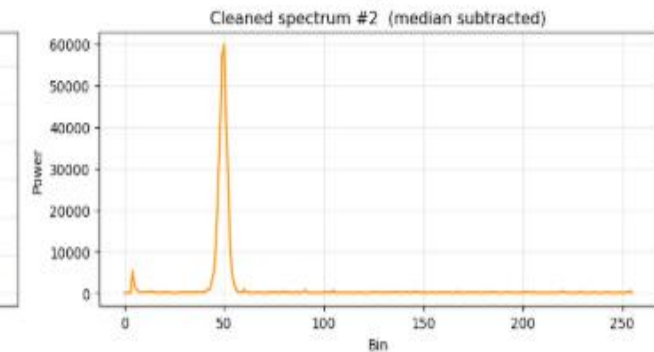
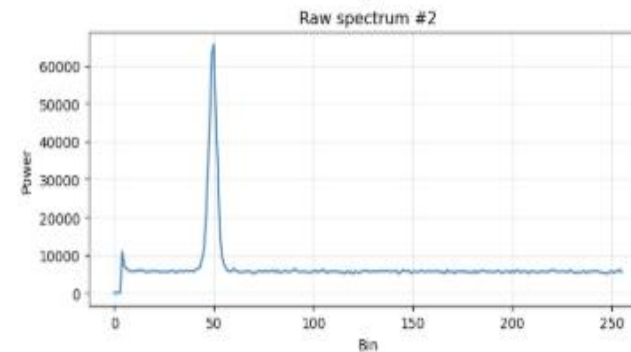
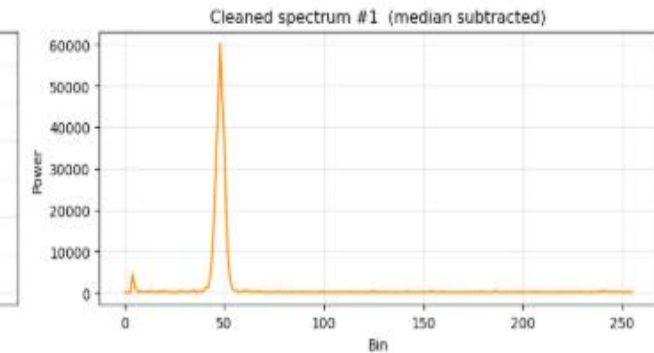
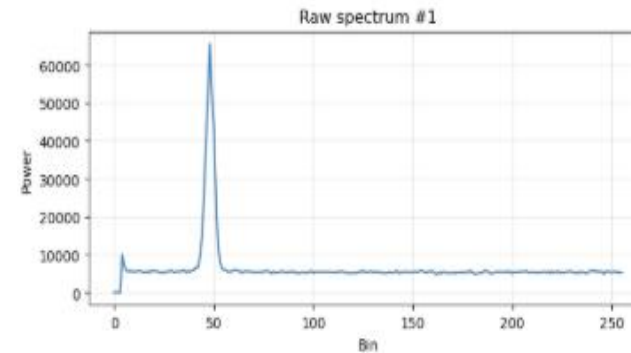
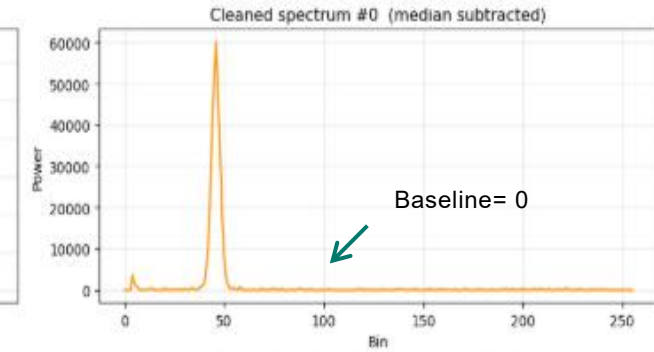
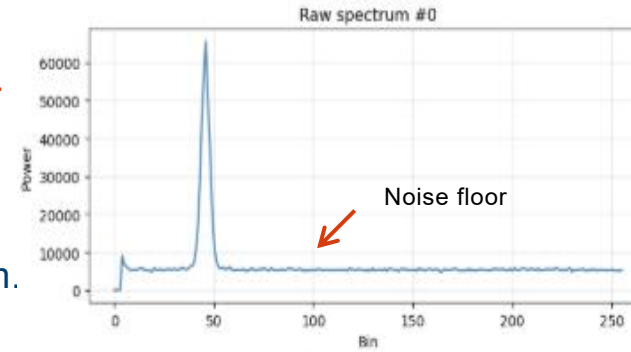
Peak only takes **10/256 bins**.

Median reflects noise floor not spike.

[What happens after Cleaning?]

Baseline sits at zero, peak is preserved.

Applied to all 312 spectra.



VELOCITY AXIS: bins to m/s

Each 256 bin = Doppler frequency = V_{los} (m/s)

Full velocity range: $0 \rightarrow \sim 38.8$ m/s across 256 bins

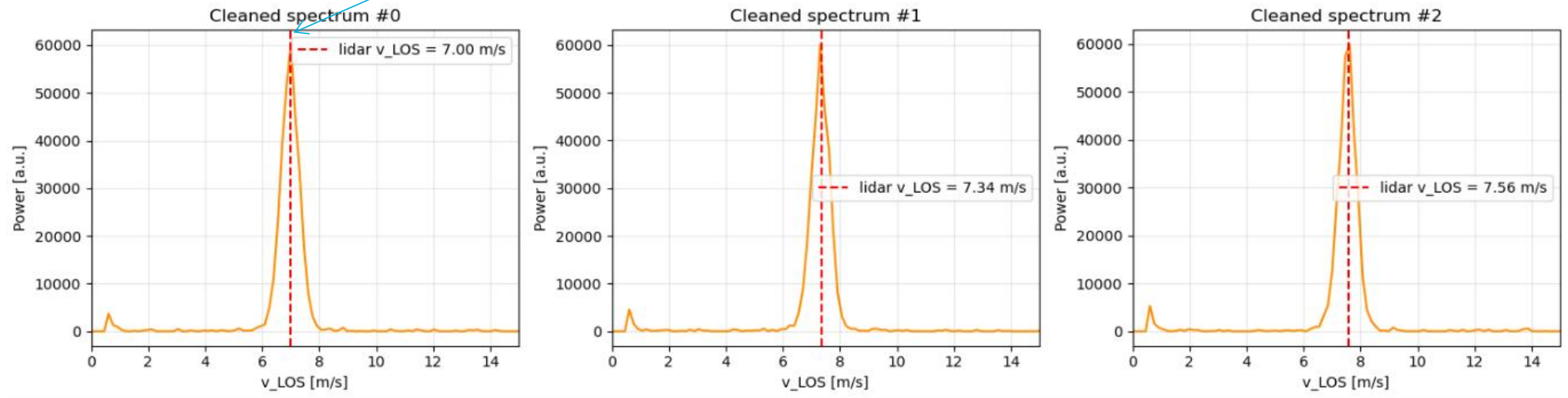
frequency resolution per bin

$$\Delta f = \frac{\text{bandwidth}}{N} = \frac{50 \times 10^6}{256} \approx 1.95 \times 10^5 \text{ Hz per bin}$$

velocity resolution per bin

$$\Delta v_{\text{los}} = \frac{1}{2} \Delta f \lambda \approx 0.152 \text{ m/s per bin}$$

peak at bin 46 $\rightarrow 46 \times 0.152 \approx 7.0$ m/s
matches lidar's report of 7.00 m/s exactly.



Bin axis is now a physical wind speed axis

Centroid peak detection

Centroid/ Integration formula applied to each 312 spectra
(Lecture 23)

- Integration in ± 15 bin window around max. Avoid noise pull from bins far from the peak.

$$f_{peak} = \frac{\int f \cdot p(f) df}{\int p(f) df} \implies v_{peak} = \frac{\sum_k v_k \cdot p_k}{\sum_k p_k}$$

Sub-bin precision (~ 0.02 m/s) vs. discrete maximum (0.152 m/s step).
No peak shape assumption.

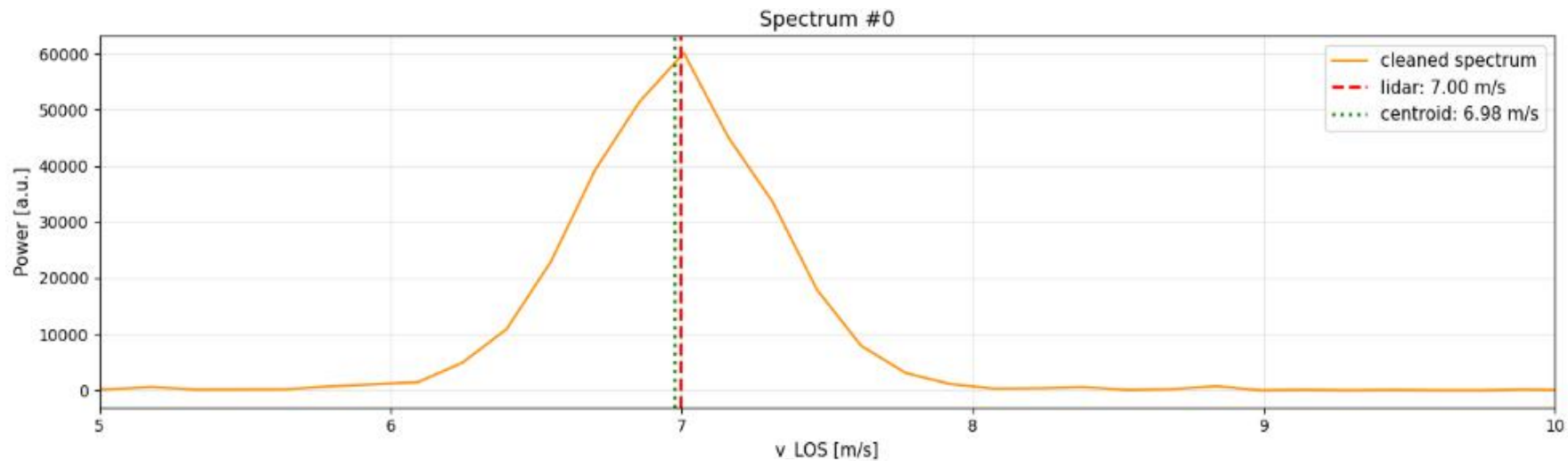
Cross-check: Gaussian fit (helper Gauss_Fit_Peak)

Mean $\Delta = +0.01$ m/s

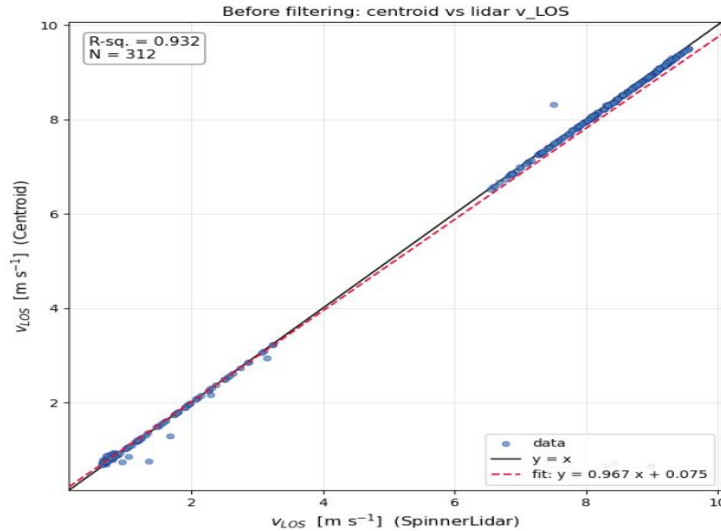
$\sigma = 0.06$ m/s

Max $\Delta = 0.28$ m/s

→ Centroid $\approx$ Gaussian



Correlation, outlier identification, filtering



Correlation before filtering

$R^2 = 0.932$, $N = 312$

Most points sit on $y = x$
→ centroid agrees with lidar.

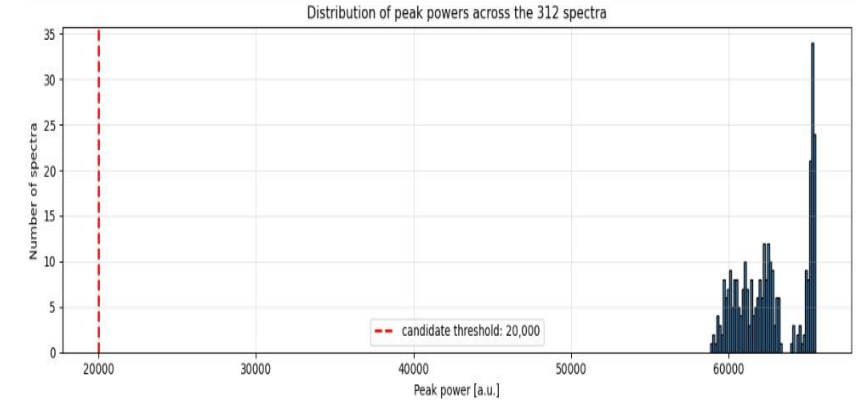
But **6 Outliers**



First attempt: Peak power distribution

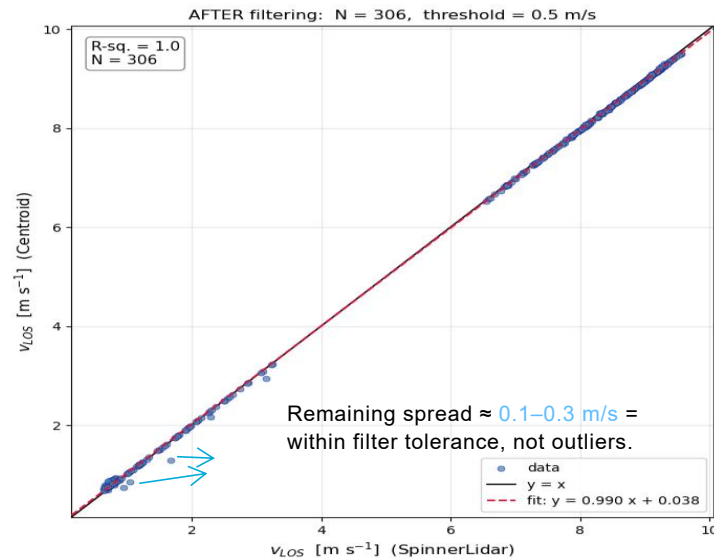
- No weak peak population
- 58k - 66k a.u (one tight cluster)
- peak-power filter **failed**
- Blade shadow doesn't weaken peak.

Need different signal → use disagreement with lidar's own v_{LOS}



[Look at what the outlier spectra actually contain]

- Every outlier spectrum has **two peaks**



Filter criterion

$$|v_{LOS,centroid} - v_{LOS,lidar}| > 0.5 \text{ m/s} \Rightarrow \text{reject}$$

Good spectra: $|\Delta| < 0.1 \text{ m/s}$

Bad spectra: $|\Delta| > 7 \text{ m/s}$

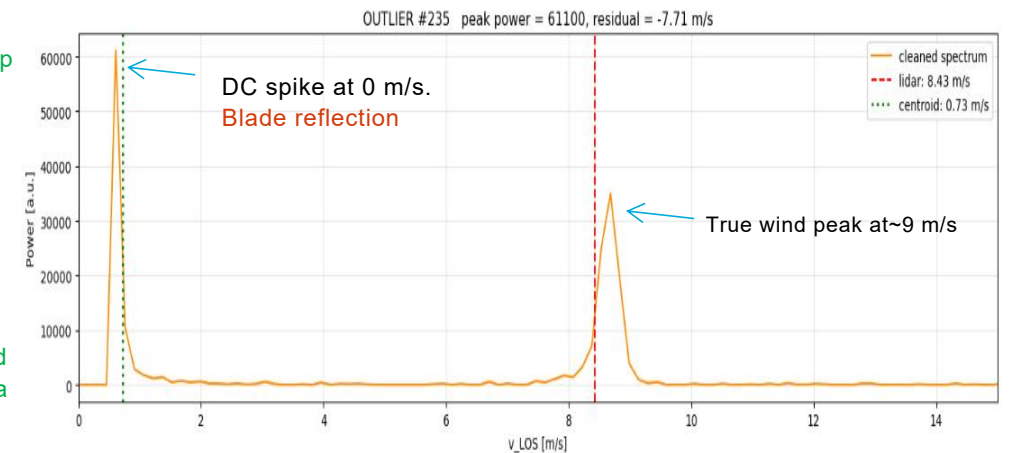
0.5 m/s sits in the empty gap



After filtering: $R^2 = 1.00$, $N = 306$ (6 rejected, 1.9%)

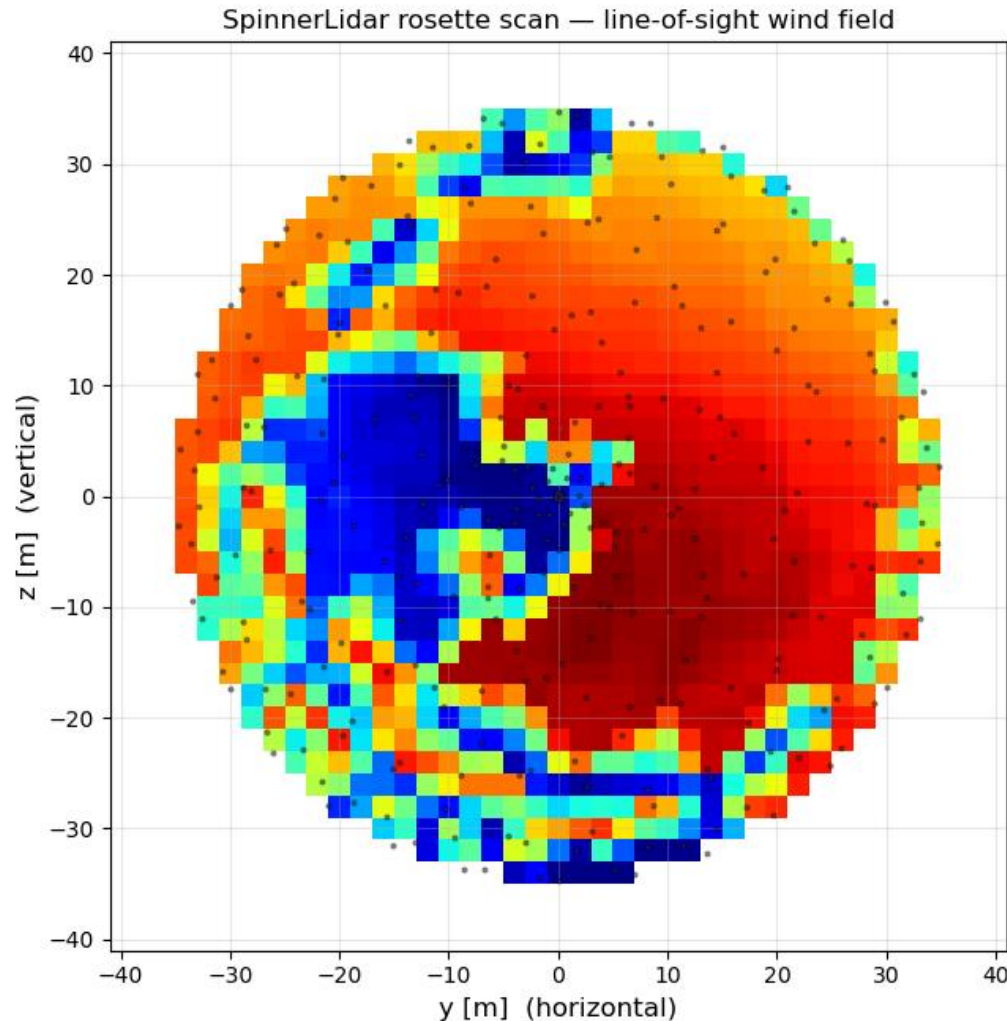
$y = 0.99x + 0.038$ →
essentially perfect $y = x$

- centroid method validated
for uncontaminated spectra



Outliers = rotor blade reflections at zero Doppler → after filtering, centroid matches lidar perfectly ($R^2 = 1.00$).

Filtered wind field



- Rosette-scan pattern samples 312 points on a circular plane **70 m upstream of the rotor, radius ≈ 35 m.**
- Free-stream region on the right **$\sim 7\text{--}9$ m/s**, undisturbed inflow.
- Low-momentum region on the left **$\sim 1\text{--}3$ m/s**, sharply bounded.
- **Wake from an upstream turbine.** Slower air stolen by a neighbour, pushed toward AV04 by the wind.
- Sharp boundary and turbulence inside = **real atmospheric structure NOT measurement error.**

The filtered SpinnerLidar captured a real wind-farm wake in the rotor inflow plane.

THANK YOU FOR LISTENING